MTH405, MTH633 Assignment 1 Solution
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Total Marks: 10 **Due Date: December 21, 2021**

Q1: Find the order of each element of the group {1, -1, i, -i} **Marks = 5**

Solution:

Let $G = \{1, -1, i, -i\}$

Because this group under multiplication so $e = 1$

By using Formula $a^n = e \quad \forall\ a \in G$ Here n is called the order of the element.

Putting $a = 1$ and $n = 1$ in the formula

$(1)^2 = 1 = e$, So, 1 is order of 1

Now, Putting $a = -1$ and $n = 2$ in the formula

$(-1)^2 = 1 = e$ So, 2 is order of -1

Similarly, Putting $a = i$ and $n = 4$ in the formula

$(-1)^4 = 1 = e$ So, 4 is order of i

And, Putting $a = --i$ and $n = 4$ in the formula

$(-i)^4 = 1 = e$ So, 4 is order of $-i$

Q2: Show that the set of integers ‹Z, +› is a group under addition. **Marks = 5**

Solution:

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Case 1:

There exists an identity element in the group that fixes every element under given the binary operation.

Yes, the number 0 is the identity since

$0 + k = k + o, \quad \forall k \in Z$

Case 2:

Closure: given any two-element *a* and *b* in the set, we need to show that *a+ b* in the set.

Yes, given any two integers *a* and *b*, there same *a + b* is again an integer.

Case 3:

Associativity:

Yes, addition in Z is associative. i.e., $(a + b) + c = a + (b + c)$

Case 4:

Existent of inverse: given any element *a* in the set, we need to find another element such that *a* and its inverse commute and their operation together given the identity element.

Yes, given $a \in G$ then -*a* is the inverse. Since $a + (-a) = (-a) + a = 0$

Since *Z* with addition satisfies above all 4 conditions. So, $< Z, + >$ is a group under addition